

Optimization

4.3.1 Exterior penalty function method (EPFM)

José Firmino Aguilar Madeira

From KKT to algorithms: how can we use Chapter 3?

- 4.2 ended like this: KKT says what a solution is, not how to find it.
- In Ch. 3 we have good methods — but for problems *without* constraints.

Question

Can we use the unconstrained methods of Ch. 3 to solve a problem **with** constraints?

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Yes — if we make violating the constraints **cost something.**

$$P(\mathbf{x}, R) = f(\mathbf{x}) + R \cdot (\text{violation})^2, \quad R > 0 \text{ the “price” of violation.}$$

P is minimized with Ch. 3, for ever larger values of R .

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P is minimized with Ch. 3, for ever larger values of R .

Increasing R makes us respect the constraints better, but makes the unconstrained problem harder.

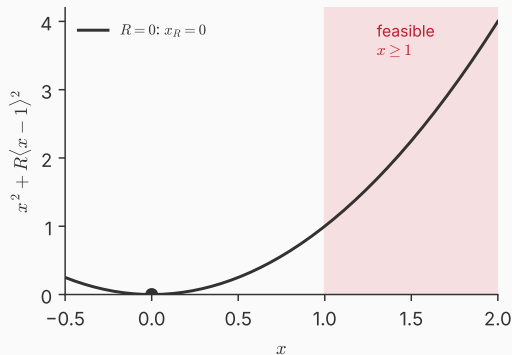
(This is the key sentence of this deck.)

Idea: pay for violations

$$\min x^2 \quad \text{s.t.} \quad g = x - 1 \geq 0.$$

Without the constraint we would want $x = 0$ — but it is forbidden. We pay for the violation:

$$P(x, R) = x^2 + R \langle x - 1 \rangle^2, \quad \langle g \rangle = \min(g, 0).$$



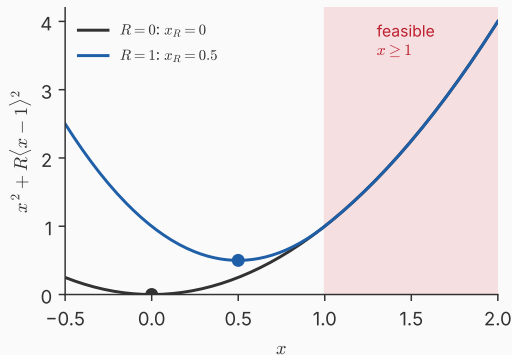
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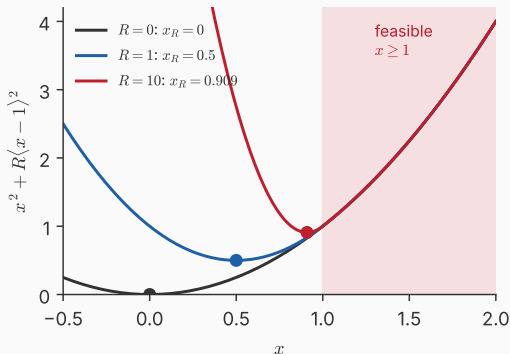
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For $x < 1$:

$$2x + 2R(x - 1) = 0 \Rightarrow x_R = R/(1 + R):$$

$$0 \rightarrow \frac{1}{2} \rightarrow 0.909 \rightarrow \cdots \rightarrow 1.$$

The “wall” gets steeper and the minimum approaches $x^* = 1$ — **always from the forbidden side.**

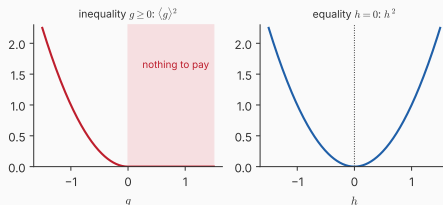


How do we measure a violation?

Inequality $g_j(\mathbf{x}) \geq 0$: there is a violation only when $g_j < 0$:

$$\langle g_j \rangle = \min(g_j, 0), \quad \text{we pay } \langle g_j \rangle^2.$$

Equality $h_\ell(\mathbf{x}) = 0$: there is violation **on both sides**: we pay h_ℓ^2 .



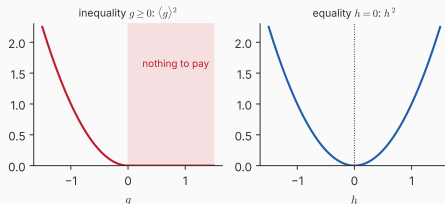
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Exterior penalty function

$$P(\mathbf{x}, R) = f(\mathbf{x}) + R \left[\sum_{j=1}^J \langle g_j(\mathbf{x}) \rangle^2 + \sum_{\ell=1}^K h_\ell(\mathbf{x})^2 \right]$$

Where the inequalities hold, **nothing is paid**; outside, one pays quadratically. An equality penalizes deviations on **both sides**.

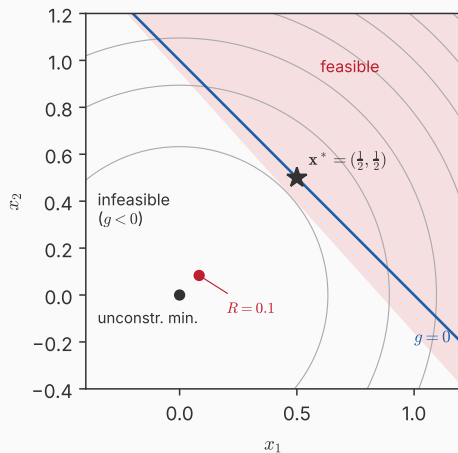
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Example: approaching from outside

$$\min x_1^2 + x_2^2 \quad \text{s.t.} \quad g = x_1 + x_2 - 1 \geq 0.$$

Solution (KKT): $\mathbf{x}^* = (\frac{1}{2}, \frac{1}{2})$, $f^* = \frac{1}{2}$, $\lambda^* = 1$.

$P = x_1^2 + x_2^2 + R \langle x_1 + x_2 - 1 \rangle^2$, for
 $R = 0.1, 1, 10, 100$.

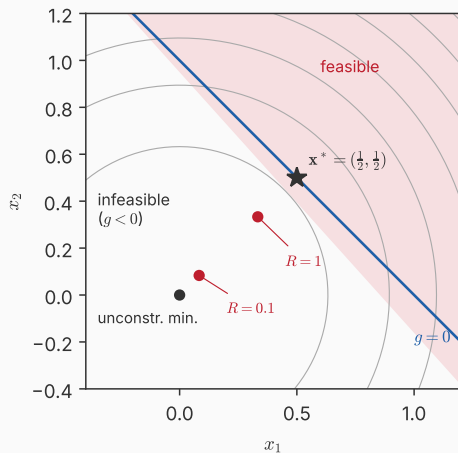


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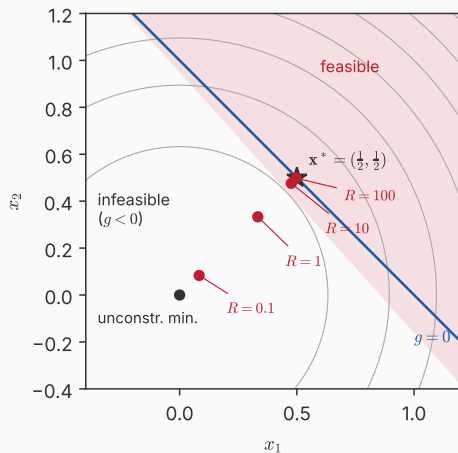


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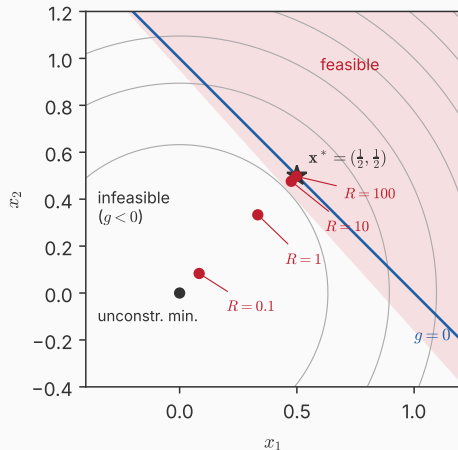
$P = x_1^2 + x_2^2 + R \langle x_1 + x_2 - 1 \rangle^2$, for
 $R = 0.1, 1, 10, 100$.

Question

Are the points $\mathbf{x}_R$ feasible?

No: all have $g(\mathbf{x}_R) < 0$. They approach $\mathbf{x}^*$ **from the infeasible side** — hence the name **exterior penalty**.

This is what happens when some constraint active at $\mathbf{x}^*$ has $\lambda^* > 0$; if the unconstrained minimum were feasible, $\mathbf{x}_R$ would be it.



The example worked out

In the violated region ($x_1 + x_2 < 1$):

$$\nabla P = \mathbf{0}: \quad 2x_1 + 2R(x_1 + x_2 - 1) = 0, \quad 2x_2 + 2R(x_1 + x_2 - 1) = 0 \Rightarrow x_1 = x_2 = s = \frac{R}{1+2R} \rightarrow \frac{1}{2}.$$

R	$\mathbf{x}_R$	$g(\mathbf{x}_R)$	$\ \mathbf{x}_R - \mathbf{x}^*\ $
1	(0.333, 0.333)	-0.333	0.236
10	(0.476, 0.476)	-0.0476	0.0337
100	(0.4975, 0.4975)	-0.0050	0.0035
1000	(0.4998, 0.4998)	-0.0005	0.00035

- **Does it work?** Yes: $\mathbf{x}_R \rightarrow \mathbf{x}^*$ and $g(\mathbf{x}_R) = -\frac{1}{1+2R} \rightarrow 0$.
- **How fast?**
 $\|\mathbf{x}_R - \mathbf{x}^*\| = \frac{1}{\sqrt{2}(1+2R)} = O(1/R)$:
each decimal place costs $R \times 10$.
- $f(\mathbf{x}_R) < f^*$: from outside, f “cheats” from below and rises to f^* .

The KKT conditions reappear in the limit

Stationarity of the penalized problem:

$$\nabla f(\mathbf{x}_R) + 2R \langle g(\mathbf{x}_R) \rangle \nabla g(\mathbf{x}_R) = \mathbf{0},$$

that is,

$$\nabla f(\mathbf{x}_R) = \underbrace{-2R \langle g(\mathbf{x}_R) \rangle}_{\lambda_R \geq 0} \nabla g(\mathbf{x}_R).$$

In the example: $\lambda_R = \frac{2R}{1+2R} \rightarrow \lambda^* = 1$.

R	1	10	100	1000
λ_R	0.667	0.952	0.995	0.9995

4.2 $\rightarrow$ 4.3.1

4.2, at the optimum: $\nabla f = \lambda^* \nabla g$.

4.3.1, during the penalization:

$\nabla f = \lambda_R \nabla g$, $\lambda_R \rightarrow \lambda^*$ (if $\mathbf{x}_R \rightarrow \mathbf{x}^*$ and LICQ at $\mathbf{x}^*$).

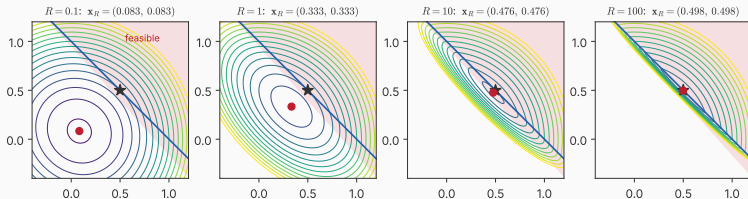
The penalty estimates the multipliers without searching for them: they come for free, from the violation.

And if the constraint is inactive?

$\min(x_1 - 1)^2 + (x_2 - 1)^2$ s.t. $9 - x_1^2 - x_2^2 \geq 0$: the unconstrained minimum $(1, 1)$ has $g = 7 > 0$.

There the penalty is zero, so $\mathbf{x}_R = (1, 1)$ for all R and $\lambda_R = 0 = \lambda^*$ — there is nothing to correct.

The price of increasing R : ill-conditioning



In the violated region, $\nabla^2 P = 2I + 2R \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$:
eigenvalues 2 and $2 + 4R$,

$$\kappa(\nabla^2 P) = 1 + 2R.$$

$R = 10^4$: $\kappa = 2 \times 10^4$ — steepest descent (3.3.1)
makes practically no progress; Newton and CG suffer
less, but finite differences become inaccurate.

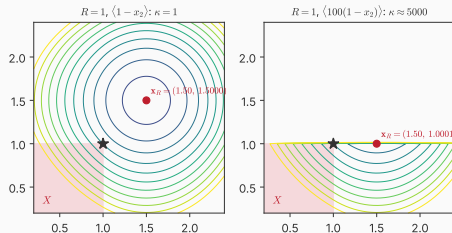
The tension of the method

more feasibility $\iff R \uparrow \iff$ worse
conditioning

The level curves become **very elongated** along
 $g = 0$.

The scale of the constraints matters

$\min (x_1 - 2)^2 + (x_2 - 2)^2$ s.t. $g_1 = 1 - x_1 \geq 0$ and $g_2 = 1 - x_2 \geq 0$ — or, the same region, $\tilde{g}_2 = 100(1 - x_2) \geq 0$; $\mathbf{x}^* = (1, 1)$.



$R \langle \tilde{g}_2 \rangle^2 = 10^4 R \langle g_2 \rangle^2$: the same “physical” violation in x_2 weighs 10^4 times more.

R	$x_{1,R}$	$x_{2,R}$ (with $\tilde{g}_2$)	$\kappa(\nabla^2 P)$
1	1.50	1.0001	5000
10	1.091	1.00001	9091
100	1.0099	1.000001	9901

One constraint almost satisfied, the other badly violated, and $\kappa \approx 10^4$ from the first R . With g_2 : $x_{1,R} = x_{2,R} = \frac{2+R}{1+R}$, $\kappa = 1$.

The method is simple, but not invariant to the scale of the constraints

$g \geq 0$ and $100g \geq 0$: same region, different penalized problem. Before penalizing, put the constraints on comparable scales (e.g. divide each g_i by a typical value).

And with equalities?

The example from 4.1: $\min 2x_1^2 + x_2^2$ s.t. $h = x_1 + x_2 - 1 = 0$.

$$P = 2x_1^2 + x_2^2 + R(x_1 + x_2 - 1)^2 \Rightarrow x_2 = 2x_1, \quad x_1 = \frac{2R}{4 + 6R}.$$

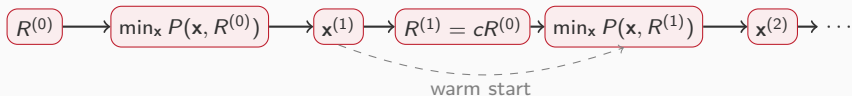
R	$\mathbf{x}_R$	$h(\mathbf{x}_R)$	$\beta_R = -2Rh$
1	(0.2000, 0.4000)	-0.4000	0.800
10	(0.3125, 0.6250)	-0.0625	1.250
100	(0.3311, 0.6623)	-0.0066	1.325
1000	(0.3331, 0.6662)	-0.0007	1.332

- $\mathbf{x}_R \rightarrow (\frac{1}{3}, \frac{2}{3})$, $h(\mathbf{x}_R) \rightarrow 0$.
- $\beta_R \rightarrow \beta^* = \frac{4}{3}$ (with $\mathcal{L} = f - \beta h$, as in 4.2; the λ^* of 4.1 is $-\frac{4}{3}$).
- $x_2 = 2x_1$ — that is, $\nabla f \parallel \nabla h$ — for every R .

The same mechanism

Inequalities: $\lambda_R = -2R \langle g \rangle \rightarrow \lambda^*$. Equalities: $\beta_R = -2Rh \rightarrow \beta^*$.

Exterior penalty algorithm



Two iteration indices: t , the outer cycle (changes R); k , the iterations of the unconstrained method of Ch. 3 within each cycle.

Exterior penalty ($\mathbf{x}^{(0)}$, $R^{(0)}$, $c > 1$, $\varepsilon_{\text{viol}}$, ε_x)

```
1: set  $t = 0$ 
2: loop
3:   set  $\mathbf{x}^{(t+1)} = \arg \min_{\mathbf{x}} P(\mathbf{x}, R^{(t)})$ , from  $\mathbf{x}^{(t)}$  ▷ Ch. 3
4:   set  $\text{viol.} = \max(\max_j |\langle g_j(\mathbf{x}^{(t+1)}) \rangle|, \max_{\ell} |h_{\ell}(\mathbf{x}^{(t+1)})|)$ 
5:   if  $\text{viol.} \leq \varepsilon_{\text{viol}}$  and  $\|\mathbf{x}^{(t+1)} - \mathbf{x}^{(t)}\| \leq \varepsilon_x \max(1, \|\mathbf{x}^{(t)}\|)$  then return
6:   end if
7:   set  $R^{(t+1)} = c R^{(t)}$ ;  $t = t + 1$ 
8: end loop
```

- The starting point **need not be feasible**.
- Equalities and inequalities are handled at the same time.
- The stopping criterion is **practical** (small violation and stalled iterates), not a universal theoretical condition.

How to choose R , warm start and cost

- As a rule of thumb, start with a **moderate** $R^{(0)}$ and increase it progressively ($R^{(t+1)} = cR^{(t)}$, $c > 1$). Starting right away with a huge R would mean solving a single, very ill-conditioned problem. (In the course examples we typically use $R^{(0)} = 0.1$ – 1 and $c = 10$.)
- **Warm start**: each minimization starts from the previous solution — the $\mathbf{x}_R$ change little from one R to the next.
- The first problems **need not be solved to high accuracy**: they will soon be replaced. Spending evaluations to solve very well a P that is about to change is wasteful (in the comparison of 4.3.2, for simplicity, all cycles with 10^{-10}).
- **Inner method**: any one from Ch. 3. Without derivatives, Nelder–Mead can be used; with `fminunc`, supply $\nabla P = \nabla f + 2R \sum \langle g_j \rangle \nabla g_j + 2R \sum h_\ell \nabla h_\ell$.
- **Cost** = sum of the calls to P in the inner minimizations (in the exterior method, each call evaluates f once: this is n_f). In the example, with Nelder–Mead: 6 cycles, about 140 evaluations per cycle, $n_f = 844$ (without the final evaluation) until $\|\mathbf{x}^{(t)} - \mathbf{x}^*\| \leq 10^{-4}$ (comparison in 4.3.2).

The price of the exterior penalty

Advantages

- Turns the problem into a sequence of unconstrained problems.
- Reuses the methods of Ch. 3.
- Does not require a feasible starting point.
- Handles equalities and inequalities.
- Estimates the multipliers (λ_R, β_R) .

Limitations

- High accuracy requires large R (error $O(1/R)$).
- Large $R \Rightarrow$ ill-conditioning ($\kappa \sim R$).
- With $R < \infty$, the solution may remain **slightly infeasible**.
- Several inner problems must be solved.
- Depends on the **scale** of the constraints.
- The iterates **leave X**: f, g_j must be computable outside it (negative thickness. . .).

We want to approximate the constraint better without letting $R \rightarrow \infty$.

Beyond the course: exact penalties (e.g. L^1) and the augmented Lagrangian seek to avoid some of these limitations.

On the computer: MATLAB implementation

The core, in MATLAB:

```
br = @(z) min(z, 0); % operator <.>
for t = 1:tmax
    P = @(x) f(x) + R*(sum(br(g(x)).^2) + sum(h(x).^2));
    [xn,~,~,out] = fminsearch(P, x, opt); % from x
    info.nfev = info.nfev + out.funcCount;
    viol = max([abs(br(g(xn))); abs(h(xn)); 0]);
    if viol <= tolviol && ...
        norm(xn - x)/max(1, norm(x)) <= tolx
        x = xn; break
    end
    x = xn; R = c*R; % increase R
end
```

The full function (PenaltyExterior, with info.nfev, info.u and info.lambda) is in the course companion code.

Call:

```
f = @(x) x(1)^2 + x(2)^2;
g = @(x) x(1) + x(2) - 1; % >= 0
h = @(x) []; % no equalities
[x, fx, info] = PenaltyExterior(f, g, h, ...
                                [0 0], 0.1, 10, 1e-6, 1e-6, 12)
info % nit = 10 (R = 1e8); u ~ 1
     % nfev ~ 130-210 per cycle
```

info.nfev counts *all* the calls to f (inner ones, via funcCount, and the final one).

The idea is just: $P = f + R \times \text{violation}^2 \rightarrow$ minimize $\rightarrow$ increase $R \rightarrow$ warm start $\rightarrow$ repeat.

Exam-style exercise (1/2): the function and the stationary points

Statement

$\min f(\mathbf{x}) = x_1^2 + 2x_2^2 \quad \text{s.t.} \quad g(\mathbf{x}) = x_1 + x_2 - 3 \geq 0.$

a) With the exterior penalty, state the function to minimize. b) Find the stationary points as functions of R and their limit as $R \rightarrow \infty$. c) With $R = 10$, perform two Nelder–Mead iterations ($\gamma = 2$, $\beta = 0.5$) from the simplex $\{(0, 0), (1, 0), (0, 1)\}$.

a) $P(\mathbf{x}, R) = x_1^2 + 2x_2^2 + R \langle x_1 + x_2 - 3 \rangle^2, \quad \langle g \rangle = \min(g, 0).$

b) If $g \geq 0$: $P = f$ and $\nabla f = \mathbf{0}$ gives $(0, 0)$; but $g(0, 0) = -3 < 0$ — rejected. If $g < 0$:

$$\frac{\partial P}{\partial x_1} = 2x_1 + 2R(x_1 + x_2 - 3) = 0, \quad \frac{\partial P}{\partial x_2} = 4x_2 + 2R(x_1 + x_2 - 3) = 0 \Rightarrow x_1 = 2x_2 \Rightarrow \mathbf{x}_R = \left(\frac{6R}{2 + 3R}, \frac{3R}{2 + 3R} \right),$$

with $g(\mathbf{x}_R) = -\frac{6}{2 + 3R} < 0$ (the case is confirmed).

$$R \rightarrow \infty: \mathbf{x}_R \rightarrow \mathbf{x}^* = (2, 1), f^* = 6; \quad \lambda_R = -2R \langle g(\mathbf{x}_R) \rangle = \frac{12R}{2 + 3R} \rightarrow \lambda^* = 4.$$

Check (KKT): $\nabla f(\mathbf{x}^*) = (4, 4) = 4 \nabla g(\mathbf{x}^*)$, with $\lambda^* = 4 \geq 0$.

Exam-style exercise (2/2): c) Nelder–Mead with $R = 10$

$P(\mathbf{x}) = x_1^2 + 2x_2^2 + 10(x_1 + x_2 - 3)^2$; from b):

$\mathbf{x}_R = (1.875, 0.9375)$, $P(\mathbf{x}_R) = 5.625$.

Start: $P(0,0) = 90$, $P(1,0) = 41$, $P(0,1) = 42$.

Iteration 1. $\mathbf{x}_l = (1, 0)$, $\mathbf{x}_g = (0, 1)$, $\mathbf{x}_h = (0, 0)$; $\mathbf{x}_c = (0.5, 0.5)$.

$\mathbf{x}_r = 2\mathbf{x}_c - \mathbf{x}_h = (1, 1)$, $P = 13 < P(\mathbf{x}_l) \Rightarrow$ **expansion**:

$\mathbf{x}_{new} = 3\mathbf{x}_c - 2\mathbf{x}_h = (1.5, 1.5)$, $P = 6.75 < 13$: accepted.

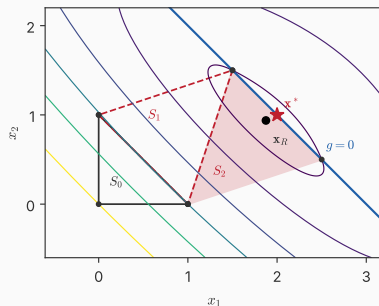
Iteration 2. $\mathbf{x}_l = (1.5, 1.5)$, $\mathbf{x}_g = (1, 0)$, $\mathbf{x}_h = (0, 1)$;

$\mathbf{x}_c = (1.25, 0.75)$.

$\mathbf{x}_r = (2.5, 0.5)$, $P = 6.75$: $P(\mathbf{x}_l) \leq P(\mathbf{x}_r) < P(\mathbf{x}_g) \Rightarrow$ **reflection** accepted.

Final simplex: $\{(1.5, 1.5), (1, 0), (2.5, 0.5)\}$, with $P = 6.75, 41, 6.75$.

The new points lie on $g = 0$ (zero penalty). The best value, 6.75, is still above $P(\mathbf{x}_R)$: the simplex would keep closing in on $\mathbf{x}_R$, which is infeasible; with a larger R , $\mathbf{x}_R$ gets closer to $\mathbf{x}^*$.



Level curves of P ($R = 10$); S_0 , S_1 , S_2 : the simplices.

KEY TAKEAWAYS — 4.3.1

1. The exterior penalty makes violation **cost**: $P = f + R \left[\sum \langle g_j \rangle^2 + \sum h_\ell^2 \right]$, minimized with Ch. 3 for increasing R (warm start).
2. Inequalities pay only when violated; equalities pay on both sides.
3. $\mathbf{x}_R \rightarrow \mathbf{x}^*$ as $R \rightarrow \infty$ (under assumptions: f, g_j, h_ℓ continuous; $\mathbf{x}_R$ global minimizers of P , with $(\mathbf{x}_R)$ bounded; $\mathbf{x}^*$ unique — without uniqueness, every limit point is a global minimum); if some active constraint has $\lambda^* > 0$, they approach it **from outside** (infeasible), with typical error $O(1/R)$.
4. The KKT conditions reappear in the limit: $\lambda_R = -2R \langle g_j \rangle \rightarrow \lambda^*$, $\beta_R = -2Rh_\ell \rightarrow \beta^*$ (if $\mathbf{x}_R \rightarrow \mathbf{x}^*$ and LICQ at $\mathbf{x}^*$).
5. **Increasing R makes us respect the constraints better, but makes the unconstrained problem harder** ($\kappa \sim R$); and the penalized problem depends on the scale of the constraints — put them on comparable scales before penalizing.

Next (4.3.2): *what if f cannot even be evaluated outside X ? And what if, instead of allowing violations and punishing them, we never left the feasible region? — barrier methods.*

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 17.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012.

Resources

- Companion code: `PenaltyExterior` (`info.nfev`, `info.u`, `info.lambda`).
- MATLAB: `fminsearch`, `fminunc` (inner minimization; `funcCount`).